ModelIt!

Student Activity Pack

# Why Pandemics Go Exponential

The Math Behind How One Infection Becomes a Global Crisis

Grade 09L2 | NGSS HS-LS2-6, HS-ETS1-4 | Systems and System Models

STUDENT NAME _____ DATE _____



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# 1 Pre-Assessment

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**What Do You Know?** Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **What do you remember about the COVID-19 pandemic? What measures were taken to slow the spread?**

2. **Why do you think some diseases spread faster than others?**

3. **What does 'exponential growth' mean, and why is it relevant to pandemics?**

4. **Draw what you think the number of infections looks like over time during an uncontrolled epidemic.**

# 2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

| Skill | Expert (4) | Proficient (3) | Developing (2) | Beginning (1) |
|---|---|---|---|---|
| **Identifying Components** | EXPERT<br>Correctly identifies and sorts ALL components as internal or external with detailed reasoning | PROFICIENT<br>Correctly identifies and sorts MOST components with some reasoning | DEVELOPING<br>Identifies components but has difficulty sorting or explaining | BEGINNING<br>Needs significant help identifying components |
| **Establishing Relationships** | EXPERT<br>Accurately describes ALL positive and negative relationships with clear evidence | PROFICIENT<br>Accurately describes MOST relationships with some evidence | DEVELOPING<br>Identifies some relationships but struggles with direction | BEGINNING<br>Needs help understanding how components connect |
| **Running Simulations** | EXPERT<br>Makes accurate predictions AND explains surprising results using evidence | PROFICIENT<br>Makes reasonable predictions and records clear observations | DEVELOPING<br>Runs simulations but predictions or observations are incomplete | BEGINNING<br>Needs help running simulations or recording data |
| **Explaining Results** | EXPERT<br>Uses multiple pieces of evidence to explain HOW and WHY systems interact | PROFICIENT<br>Uses evidence to explain system interactions | DEVELOPING<br>Gives partial explanations with limited evidence | BEGINNING<br>Needs help connecting evidence to explanations |

# 3 Component Analysis

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### What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

### Sort These Components

Susceptible Population | Infected Population | Recovered Population | Contact Rate | Transmission Rate | Vaccination Rate | Quarantine Effectiveness

| EXTERNAL | INTERNAL |
|---|---|
| | |

**Explain your reasoning: Why did you sort each component the way you did?**

# 4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

**? Explain ONE relationship in your model. What happens when one component changes?**

# 5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

## 1 Scenario 1: Uncontrolled Epidemic

High contact rate, no interventions

MY PREDICTION

WHAT I OBSERVED

## 2 Scenario 2: Early vs. Late Intervention

Quarantine at 1% infected vs. 10% infected

MY PREDICTION

WHAT I OBSERVED

## 3 Scenario 3: Vaccination Effect

Moderate vaccination rate throughout epidemic

MY PREDICTION

WHAT I OBSERVED

# 6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

## Possible New Components

**Hospital Capacity** — The maximum number of patients the healthcare system can treat simultaneously — when exceeded, mortality increases dramatically

**Asymptomatic Transmission Rate** — The percentage of infected people who show no symptoms but can still transmit the disease, making detection and isolation much harder

**Reinfection Rate** — The rate at which recovered individuals lose immunity and become susceptible again, turning the SIR model into an SIRS model with no permanent recovery

## My Extension

**A Which new component would you add to the model?**

**B How would it connect to the existing components? (Draw or describe)**

**C What evidence or reasoning supports your connection?**

# 7 Reflection Questions

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**The Big Question**

**How does one person's infection become a global pandemic in weeks?** Use evidence from your simulation to support your answers.

1. **Why does the epidemic curve naturally peak and decline even without intervention? What does this tell you about the SIR model's built-in balancing feedback?**

2. **Your model shows that the same quarantine implemented at 1% infection vs. 10% produces dramatically different outcomes. Why does timing matter so much in exponential systems?**

3. **How does vaccination change the dynamics compared to quarantine? Which intervention affects which component of the SIR model?**

4. **What real-world factors make pandemic modeling more complicated than the basic SIR model?**

5. **If you were a public health official during the early days of an outbreak, what would you need to know to make good decisions?**

# 8 STEM Challenge: Design a Pandemic Response Decision Tool

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## YOUR ENGINEERING MISSION

Using your SIR model data, design a decision support tool that helps public health officials determine WHEN and HOW AGGRESSIVELY to implement interventions during an outbreak, showing the cost of delayed action.

### THE SCENARIO

A county health department needs a tool that shows, in real time, where they are on the epidemic curve and what different intervention strategies would achieve. Your team will build this tool using your model data, emphasizing the critical importance of intervention timing.

### Engineering Design Thinking

1. HOW WILL YOUR TOOL VISUALIZE THE SIR COMPARTMENTS IN REAL TIME AS THE EPIDEMIC PROGRESSES?
2. HOW WILL YOU SHOW THE PROJECTED OUTCOMES OF DIFFERENT INTERVENTION STRATEGIES SIDE BY SIDE?
3. HOW WILL YOU COMMUNICATE UNCERTAINTY — THE FACT THAT EARLY IN AN EPIDEMIC, WE DON'T KNOW EXACT $R_0$ VALUES?
4. HOW WILL YOU HELP OFFICIALS WEIGH HEALTH OUTCOMES AGAINST ECONOMIC AND SOCIAL COSTS OF INTERVENTIONS?

### My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

**! Explain your design using scientific evidence from the simulation:**

### Real Career Connection

Epidemiologists and Public Health Modelers study disease transmission patterns and design intervention strategies. They work at the CDC, WHO, state health departments, and academic research centers, earning $75,000–$150,000/year. During the COVID-19 pandemic, their models directly informed decisions affecting billions of people.